

22	In bright light, a light-dependent resistor (LDR) has a resistance of R. It is connected in series with an ideal diode and a fixed resistor of resistance R. An ideal diode has zero resistance in the forward direction and infinite resistance in the reverse direction. In which arrangement will the potential at X increase when the circuit is moved to a darker environment?			
	A	B	C	D

Answer: A

Diodes in options B and D are in reverse biased connection (like an open circuit where the diode is).

- No current flows → zero p.d. across the resistance → potential at $X = 0\text{ V}$ in both bright and dark conditions, i.e. no change in potential at X for options B and D.
- Eliminate options B and D.

Diodes in options A and C are in forward biased connection (like zero resistance where the diode is).

- Current flows → non-zero p.d. across the resistance.

Since LDR's resistance increases when moved into the dark, by Potential Divider Principle, the p.d. across the LDR will increase.

Hence in option A potential at X will increase, while in option C potential at X will decrease.

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A battery B , a variable resistor W and a uniform resistance wire PQ are connected in series. A